# CPT205 Coursework 1 – 2D Modelling Project

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# 1. Creating a Geometry

In the overall presentation of the card, there are day mode and night mode. Most elements remain the same, but the background turns black, the grass changes from light green to dark green, the sun disappears, and the stars twinkle in the night sky.

I often make birthday cards using basic shapes (such as circles, triangles, and rectangles) and complex shapes (such as heart-shaped balloons and water-drop candles) by combining these basic shapes. I also used Angle calculations to make flowers and various star shapes while adding text and lines. Implement rectangle and circle drawing functions drawRectangle and drawCircle before drawing different elements, which simplifies the use of basic graphics and improves the maintainability of the code.

## 1.1 Grass

The grass is a simple rectangle, located at the bottom of the screen, drawn using the drawRectangle function.

## 1.2 Cake

The cake consists of a cake body, cream, candle and candle flame. The cake and cream are drawn using the drawRectangle function, with the three rectangles gradually decreasing from bottom to top and the color changing from brown to light. The candles are from left to right, with the central candle being the highest. The candle flame is formed using a combination of triangles and circles, and the color is light pink.

## 1.3 Cloud

The outline of the cloud consists of an ellipse and four circles of different sizes. Draw each circle using the drawCircle function and color it white. By adjusting the center of the circle and the offset, beautiful clouds can be formed.

Add a smile, blush, and eyes to the cloud to make it more vivid. The eyes and blush are drawn using drawcircles, and the smile is represented by an arc.

## 1.4 Heart-shaped balloons

The heart-shaped balloon consists of a balloon and two strings. The heart-shaped part of the balloon consists of a triangle and two circles. By adjusting the center of the circle and the vertex position of the triangle, a realistic heart shape can be made.

## 1.5 Butterfly

A butterfly consists of an antenna, a body and wings. Each section consists of circles of different radii and shapes, drawn using the drawCircle function.

## 1.6 Flowers

A flower consists of a heart and petals. Use the drawCircle function to generate a centerpiece. The five petals rotate 72 degrees around the center.

## 1.7 Sun

The sun consists of a center and light rays. Draw the center of the Sun using the drawCircle function, with ten orange rays rotating 36 degrees around the center.

## 1.8 Birthday wishes

In order to animate the characters up and down in the future, this step needs to generate each character individually and set their initial coordinates and colors.

### 1.9 Birthday Information "Forever 18"

Generated by traversing each character in the string.

### 1.10 Stars

Stars are divided into five-pointed and four-pointed stars, which are drawn by calculating the positions of the outer and inner points, resulting in two yellow stars.

## 2. Transformation

### 2.1 Color switch for day and night modes

The initial background color is set to light blue and the grass color to light green. When you right-click, the background turns black and the grass turns dark green.

### 2.2 The disappearance of the sun and the appearance of stars

Added Boolean variable drawSunFlag to control the drawing of the sun, initially true. When you right-click, set to false, the sun disappears, and add drawStarFlag to control the display of stars. The initial state is false, and the stars appear after you right-click.

### 2.3 The candle flame is extinguished

Control the opacity of the candle flame by adjusting the flameOpacity variable, initially to 1.0, and reducing it by 0.1 with each left-key click. After 10 clicks, the flame is completely extinguished, simulating the effect of blowing out a candle.

### 2.4 Various transformations of butterflies

Set initial transformation parameters for the butterfly, including translation, rotation, and scaling. These parameters can be changed by pressing different keys, causing the butterfly to move, rotate and scale in different directions.

### 2.5 Appearance of birthday wishes

Initialize the blessing message at the top of the window, set its ordinate to 400 when you press Enter, and display it above the cake as "Forever 18".

## 3. Viewing

The display function in my code is responsible for rendering the scene in an OpenGL window. First, it sets the 2D orthogonal projection and clears the background. Next, multiple elements were drawn, including grass, sun, flowers and balloons. The grass is drawn by the drawGrass() function, and the positions of the sun and flowers are determined through a loop. The balloon is drawn according to its Y coordinate to determine whether it is in the screen.

In addition, the code draws clouds and stars, using drawCloud and the drawStar1 and drawStar2 functions. The cake is drawn by the drawCake function, and the flame is drawn by calling drawFlame several times. The character happy birthday is also drawn through a loop, with the position adjusted according to the jump offset. Finally, the birthday wishes are displayed through the drawMessage function. glFlush() is used at the end of the function to

ensure that all commands are executed, and the buffer is swapped by glutSwapBuffers() for smooth rendering and dynamic updating of the scene.

# 4. Animation

## 4.1 Release balloons

When setting the relevant variables for the balloon, you can add the Boolean variable for balloonMoving, with an initial state of false. When pressing the space bar on the keyboard, balloonMoving changes to true, the vertical axis of the balloon begins to increase as the timer function updateBalloon proceeds, and the balloon begins to rise until it flies off the screen.

## 4.2 Rotation of petals

Add a variable to the Angle of the petal, angleOffset, which increases by 5 each time the timer function updates angleOffset, causing the petal to rotate.

## 4.3 Character jumping

For each character, add jumpOffset, upperLimit, lowerLimit, move directions, update the offset according to the direction, change the direction when moving to the highest or lowest point.

## 4.4 Twinkle of stars

Add a variable isStarVisible for stars to control how visible the stars are in night mode. Each time the timer function is called, the visible state of the stars will be switched so that they will shine in the night.

# 5. Interactions

## 5.1 Butterfly Transformation (Keyboard Interaction)

- Press the lowercase letter ‘w’ key on the keyboard, the butterfly moves upward;
- Press the lowercase letter ‘s’ key on the keyboard, and the butterfly moves down;
- Press the lowercase letter ‘a’ key on the keyboard, and the butterfly moves to the left;
- Press the lowercase ‘d’ key on the keyboard, and the butterfly moves to the right;
- Press the lowercase letter ‘r’ key on the keyboard, the butterfly rotates counterclockwise;
- Press the capital letter ‘R’ key on the keyboard, and the butterfly rotates clockwise;
- Press the lowercase ‘q’ key on the keyboard to enlarge the butterfly;
- Press the lowercase letter ‘e’ key on the keyboard, the butterfly shrinks;

## 5.2 Extinguishing of candle flame (mouse interaction)

With each press of the left mouse button, the candle flame diminishes somewhat. After clicking the left mouse button ten times, the candle flame is entirely extinguished.

## 5.3 Balloon Release (Keyboard Interaction)

Upon pressing the space bar, all balloons will ascend until they exit the screen.

## 5.4 Birthday Salutations (Keyboard Interaction)

Upon pressing the Enter key, the message "Forever 18" will promptly manifest at the top of the cake.




## 5.5 Card day and night mode switch (Mouse Interaction)

Upon pressing the right mouse button, the card will transition to dark night mode; the light blue sky will transform to black, the light green grass will shift to dark green, the sun will vanish, and stars will emerge in place of the sun.

